Sophie Mirabai Woodward

PhD Candidate in Biostatistics
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Research Summary

I develop statistical methods for causal inference in spatial data settings, with a focus on addressing unmeasured spatial confounding in environmental health studies. My work integrates ideas from spatial statistics, causal inference, and semiparametric estimation to improve the validity of exposure–health effect estimation. I am also studying how foundation model embeddings can be used to borrow information across related treatments to estimate causal effects of new or rare treatments. I am broadly motivated by scientific problems that combine methodological innovation with real-world impact in health, environment, and policy.

Education

Harvard University
PhD in Biostatistics (*Expected 2027*), *AM in Biostatistics* (*2024*) *2022–2027*
Developing methods to address unmeasured spatial confounding in causal inference under the guidance of Dr. Francesca Dominici.
Selected Coursework: Probability Theory II, Statistical Inference II, Causal Inference: Theory and Practice, Epidemiology, Bayesian Methodology, Multivariate and Longitudinal Data, Social and Biological Networks (GPA 4.0/4.0).

Harvard College
BA in Statistics and Mathematics *2018–2022*
Graduated Magna Cum Laude with Highest Honors.
Selected Coursework: Probability I (PhD-level), Statistics I (PhD-level), Computer Science, Linear Models, Spatial Statistics, Generalized Linear Models, Stochastic Processes, Design of Experimental and Non-Experimental Studies, Causal Inference (GPA 3.96/4.0).

Publications and Preprints

Woodward, S.M., Kitch, J., Tec, M., Dominici, F., & Battiloro, C. (2026). Few-Shot Causal Effect Estimation for New Continuous Treatments via Foundation Model Embeddings. *Under review*.

Woodward, S.M., Sprague, N.L., Black-Ingersoll, F., Burrows, K., Amaya, F., Davis, Z., Echevarría-Ramos, M., Hass-Mitchell, T., White, K., James, P., Rodriguez, T., Oblath, R., & Nori-Sarma, A. (2026). Spatial Variation in the Association Between Extreme Heat and Psychiatric Emergency Department Visits in Greater Boston: Evidence to Inform Prevention and Policy. *Submitted*.

Woodward, S.M., Dominici, F., & Zubizarreta, J. (2025). Understanding Spatial Regression Models from a Weighting Perspective in an Observational Study of the Impact of Superfund Cleanups on Birth Outcomes. *Under review*.

Woodward, S.M., Tec, M., & Dominici, F. (2024). An Instrumental Variables Framework to Unite Spatial Confounding Methods. *Under review*.

Tec, M., Trisovic, A., Audirac, M., **Woodward, S.M.**, Hu, J., Khoshnevis, N. & Dominici, F. (2024). SpaCE: The Spatial Confounding Environment. *International Conference on Learning Representations*.

Woodward, S.M., Mork, D., Wu, X., Hou, Z., Braun, D., & Dominici, F. (2023). Combining aggregate and individual-level data to estimate individual-level associations between air pollution and COVID-19 mortality in the United States. *PLOS Global Public Health*, 3(8), e0002178.

Woodward, S.M., & Wu, X. (2023). Review 1: "Long-term Outdoor Air Pollution and COVID-19 Mortality in London: an Individual-level Analysis". *Rapid Reviews Infectious Diseases*.

Field, R. D., Moelis, N., Salzman, J., Bax, A., Ausiello, D., **Woodward, S.M.**, & Edwards, D. A. (2021). Inhaled water and salt suppress respiratory droplet generation and COVID-19 incidence and death on US coastlines. *Molecular Frontiers Journal*, 5(01n02), 17-29.

Professional Experience

2026: Associate Intern in Health Economics and Outcomes Research, Analysis Group, Summer 2026.

Awards

2025: Tom Ten Have Award at the *American Causal Inference Conference* for poster on spatial regression from a weighting perspective.

2024 and 2025: Certificate of Distinction in Teaching, Harvard Biostatistics; teaching evaluations average above 4.8/5.

2024: Selected Student Speaker at the *Accelerate: Data for Social Impact Conference*.

2022: National Science Foundation Graduate Research Fellowship.

2020: John Harvard Scholarship (GPA in the top 5% of class).

Teaching

Statistical Inference I (BST 231) — **Harvard University**
Teaching Fellow — *Spring 2024 & 2025*
Led weekly lab sections, graded assignments, and held office hours for master's and doctoral students.
Supervisor: Dr. Rui Wang.

StatStart Summer Program — **Harvard University**
Instructor — *Summer 2024 & 2025*
Volunteered to teach introductory courses on statistics, R programming, and data analysis to high school students in a computational summer program.

Introduction to Probability and Statistics — **Harvard University**
Teaching Assistant — *2020–2022*
Led weekly lab sections, graded assignments, and held office hours for undergraduate students.
Supervisor: Dr. Joseph Blitzstein.

Presentations

2026: American Causal Inference Conference, Salt Lake City, UT. Spotlight presentation, May 2026.

2026: Harvard Data Science Initiative Causal Inference Showcase, Cambridge, MA. Oral presentation, February 2026.

2025: American Causal Inference Conference, Detroit, MI. Poster presentation, May 2025.

2025: Eastern North American Region International Biometric Society Spring Meeting, New Orleans, LA. Oral presentation, March 2025.

2024: Accelerate: Data for Social Impact Conference. Oral presentation, June 2024.

2024: American Causal Inference Conference, Seattle, WA. Poster presentation, May 2024.

2022: National Collegiate Research Conference, Cambridge, MA. Poster presentation, January 2022.

Skills

Programming: R (SuperLearner, sf, spdep, mgcv, data.table, tidyverse), Python (numpy, pandas, scikit-learn, statsmodels, tensorflow, pytorch)

Service

Webinar Presenter: *Spatial Confounding and Interference in Environmental Health Studies*. CAFE Research Coordinating Center, February 2026.

Peer Reviewer: NeurIPS 2025 Workshop CauScien, Uncovering Causality in Science.

Peer Reviewer: Journal of Causal Inference.